

Jagadeep Reddy

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SUMMARY

AI Engineer transitioning from 2 years of production backend engineering at ANZ (Asia-Pacific's largest banks). Hands-on expertise building production-grade GenAI systems — hybrid RAG pipelines with measurable RAGAS evals, autonomous multi-agent systems with LangGraph, fine-tuned LLMs (LoRA/QLoRA/DPO), and real-time ML inference under strict latency budgets. Backend engineering background provides a strong foundation for reliable, production-ready AI deployments.

TECHNICAL SKILLS

Languages: Python, Java, SQL

GenAI & LLMs: RAG, LangChain, LangGraph, Fine-tuning (LoRA/QLoRA/DPO), Prompt Engineering, RAGAS, OpenAI API

Models: Mistral, Llama, BERT/roBERTa, HuggingFace TRL, vLLM, BGE-M3

Retrieval: FAISS, Qdrant, BM25, Reciprocal Rank Fusion, Cross-encoder Reranking, pgvector

Deep Learning: PyTorch, TensorFlow, Transformers

Cloud & Infra: AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes, CI/CD, FastAPI, Flask, Redis, Kafka

Data: pandas, NumPy, matplotlib, seaborn, XGBoost, SHAP, Optuna, feature engineering

Monitoring: Datadog, LangSmith

PROJECTS

IPL AI Intelligence Platform | *LangGraph, Hybrid RAG, XGBoost, FastAPI, Qdrant, PostgreSQL, Redis, RAGAS, WebSocket* | [GitHub](#)

- Production-grade multi-agent AI analytics over 18 seasons / 1.2M ball-by-ball IPL deliveries with real-time ML inference under a 300ms budget.
- Built 5-agent LangGraph orchestration (StatsQA, NarrativeQA, Prediction, Matchup, TeamVsTeam) with intent-based routing — each agent uses the right backend (SQL, RAG, or XGBoost+SHAP) rather than forcing all queries through a single LLM call.
- Hybrid retrieval: BGE-M3 dense (Qdrant HNSW) + BM25 sparse → RRF fusion → cross-encoder rerank (top-8) — RAGAS faithfulness 0.71 → 0.88 on 200-question golden eval set; CI gate blocks deploy if score drops below threshold.
- XGBoost win-probability model (AUC 0.72, Optuna-tuned) with SHAP explanations in natural language; intent classifier (71% acc, 4-class) routes live ball events in ~20ms.
- WebSocket real-time path: BallEvent → XGBoost intent (~20ms) + RAG context (~50ms) → GPT-4o-mini commentary — <300ms end-to-end, load-tested to 200 concurrent users (Locust, 60s sustained).

Production RAG System & Evaluation | *FAISS, BM25, LangChain, RAGAS, LangSmith, Cross-encoders, CI/CD* | [GitHub](#)

- Enterprise Q&A over SEC 10-K filings with a full retrieval evaluation framework covering dense, sparse, and hybrid retrieval strategies.
- Implemented three chunking strategies (fixed-size, semantic, hierarchical parent-child) to benchmark retrieval quality; hierarchical chunking improved context precision by retaining full parent context during generation.
- Built hybrid dense (FAISS IVF) + sparse (BM25/Rank-BM25) retrieval with RRF fusion and cross-encoder reranking (ms-marco-MiniLM-L-6-v2); enforced chain-of-thought prompting with strict citation injection to reduce hallucinations.
- Self-consistency hallucination detection: 3 parallel LLM responses at temperature 0.4 with automated flag on factual deviation; agentic multi-hop query decomposition with parallel sub-query retrieval and synthesis.
- RAGAS metrics (faithfulness, answer relevancy, context precision/recall) tied to GitHub Actions CI/CD gate — fails if faithfulness < 0.75; 40% improvement vs naive baseline on 500-question SEC eval set.
- Full observability via LangSmith tracing across all LangChain execution blocks to monitor latency and pinpoint retrieval/generation bottlenecks.

EXPERIENCE

AI Engineer — Independent (Self-directed upskilling)

Jul 2024 – Present

- Built 3 production-grade AI systems covering multi-agent orchestration, hybrid RAG, and real-time ML inference — all with live demos and full RAGAS eval pipelines.
- Completed PG Program in Data Science & ML (UT Austin × Great Learning); earned Andrew Ng ML Specialization, Claude 101 (Anthropic), and AI Agents for Enterprise certifications.

Software Engineer — Core Banking Platform | ANZ

Jun 2022 – Jun 2024 · Bengaluru

- Built and deployed LoanIQ APIs (Inquiry APIs + Scripted Batch) in Spring Boot handling accrual cycles for large loan portfolios; used HashMap/ArrayList for efficient RID storage and JDBC for real-time queries against high-volume financial data.
- Designed and deployed Customer Onboarding modules using Microservices architecture and Kafka message queues; achieved 90% unit test coverage ensuring production reliability at enterprise scale.
- Managed Oracle database objects using complex SQL — multi-join queries, correlated subqueries, stored procedures, and views — to serve diverse reporting needs across risk and loan teams.
- Led batch multilegging for LoanIQ 7.6.4.0 upgrade; created and deployed components across environments using UDeploy, ensuring zero-downtime rollout across production systems.

EDUCATION

Post Graduate Program in Data Science & ML — UT Austin × Great Learning (online) May 2025 – Jun 2026

BE, Information Science and Engineering — BMS Institute of Technology and Management, Bengaluru 2018 – 2022 · 8.50 CGPA

CERTIFICATIONS

Machine Learning Specialization — Andrew Ng, Coursera · Building AI Agents for Enterprise — Beyond Vectors AI · Claude 101 — Anthropic